THE OTHER ROADBLOCKS

The Mumbai-Ahmedabad corridor had been designed around Japan's Shinkansen technology, leaving Japanese behemoths Kawasaki and Hitachi as the only companies capable of supplying the E5 Shinkansen trainsets. "Effectively, it was turning out to be a single tender-like situation for us, which, ideally, is a no-no," says a senior railways ministry official, on condition of anonymity. "After a lot of discussions, we even agreed in principle to go with a limited number of vendors as it was a bilateral agreement. But the prices they quoted were atrocious, knowing we couldn't go anywhere else."

Officials say the quoted price worked out to Rs 40-60 crore per coach—taking the cost of a single trainset to around Rs 1,000 crore, several times the cost of a Vande Bharat rake.



"The first section of the Mumbai–Ahmedabad High Speed Rail Corridor will open in August 2027. We are now constructing about 15-16 km every month. The progress is very good"

ASHWINI VAISHNAW, *Union Minister of Railways*

"When Vande Bharat, which can run at 180 kmph, costs Rs 100-120 crore, a bullet train to cost Rs 1,000 crore was just not going to be justifiable in India," says Shubhranshu, a rail mobility expert who was among the lead architects of Vande Bharat. "For a project to truly symbolise India's progress, the trains had to be manufactured domestically."

By now, delays—first over land acquisition and later procurement—had also begun to inflate the project's cost. Already revised to Rs 1.08 lakh crore, the expense was heading towards Rs 1.6 lakh crore. Since repayments are denominated in Yen, exchange-rate movements further increased India's long-term liability, taking the effective cost close to Rs 2 lakh crore.

The original 2023 completion target had long slipped. Even conservative estimates placed completion of the full corridor around 2029. India proposed waiting for Japan's next-generation E10 Shinkansen instead of procuring the ageing E5 series trainsets. But the E10 is still under development and is unlikely to be available for export before the early 2030s, making that an unviable option.

If rolling stock presented one challenge, signalling posed an even bigger one. Japan's HSR network uses its proprietary Digital Signalling-Automatic Train Control (DS-ATC) system, while much of the rest of the world has standardised around the European Train Control System (ETCS). ETCS also underpins Kavach, India's automatic train protection system, making it more interoperable and economical.

Yet, even as late as last year, the project appeared to have reached the point of no return on its technology choices. The August 2025 India–Japan Annual Summit Joint Statement seemed to reaffirm that trajectory. "They (the two PMs) concurred to work towards the commencement of operations at the earliest and cooperate on the introduction of the latest Japanese Shinkansen technology in India," it said. "The Indian side appreciates Japan's offer to introduce, in the early 2030s, the E10 series of the Shinkansen that runs on the Japanese signalling system."

This was even as India had, almost simultaneously, decided on a technological pivot—adopting ETCS Level 2 in place of DS-ATC after prolonged diplomatic negotiations with Japan. The German multinational conglomerate Siemens won the Rs 4,100-crore signalling contract in May 2025. "India is making so many HSR corridors, so it is only natural that it would choose a technology that gives it the maximum flexibility in terms of choosing the core technology," says Tilak Raj Seth, head of Mobility at Siemens India.

The trains themselves were the final piece of the puzzle to fall into place this year. In February, a newly formed partnership between defence PSU Bharat Earth Movers Limited (BEML) and Hyderabad-based Medha Servo Drives—the supplier of propulsion and control systems for the first Vande Bharat trains—won the contract to build 16 trainsets for the HSR corridor. Awarded on a single-tender nomination basis, the contract is understood to carry a quoted price of about Rs 250 crore per trainset. Under the arrangement, BEML will build the coaches, undertake final assembly and fit out the interiors, while Medha will supply the propulsion, traction and electrical systems. The order builds on a 2024 contract under



NEARLY THERE
Construction advances on a bridge across the Sabarmati river for the bullet train project, Ahmedabad, Apr. 27